

Task 1 - Post-Launch Health, Incident Command & Sprint Planning

Phase 3 | Scale, Intelligence & Enterprise Readiness | Data Analyst | Sprint A - Scale & Reliability

Source Decision

The brief requires a launch performance report on real data and explicitly says production data must be used, never synthetic. No PlaceMux production dataset is supplied, so this report uses the real UCI Online Retail dataset as a documented external-data proxy. It must not be described as PlaceMux production telemetry.

Source: UCI Machine Learning Repository - Online Retail, DOI 10.24432/C5BW33. The dataset contains 541,909 transaction lines with invoice, product, date, quantity, unit price, customer and country fields.

Definition of Done Mapping

Required output	Delivered
Launch performance report	Commercial flow proxy, revenue baseline, repeat-purchase continuity and quality controls
Ranked biggest problems	Row-volume ranking with decision priorities and explicit overlap caveat
Phase-3 analytics backlog	Priority-ranked work items tied to decisions
Demo / traceability	Python source, machine-readable outputs and self-contained dashboard

1. Verified Performance Baseline

Metric	Value
Raw transaction lines	541,909
Valid purchase lines	397,884
Valid-row retention	73.4%
Clean invoices	18,532
Identified customers	4,338
Products	3,665
Countries	37
Retained gross revenue	£8,911,407.90
Mean invoice revenue	£480.87
Repeat-purchase continuity proxy	76.63% (14,194 invoices followed by a later invoice)

The retention metric is deliberately labelled a transaction-level proxy. The source does not contain a PlaceMux lifecycle, so no PlaceMux application retention rate is fabricated. The implementation can compute the proxy directly from the source and should be replaced with a production cohort definition once PlaceMux retention events exist.

Uncertainty is handled in the code with a fixed-seed bootstrap 95% confidence interval for mean invoice revenue. Revenue is also reconciled from line-level Quantity x UnitPrice to invoice aggregation.

2. Data Quality Controls

Issue	Rows	Share
Missing CustomerID	135,080	24.93%
Quantity <= 0	10,624	1.96%
Cancellation invoice lines	9,288	1.71%
UnitPrice <= 0	2,517	0.46%
Missing Description	1,454	0.27%

Exception classes can overlap; they are not additive. This prevents double-counting in the problem ranking.

3. Ranked Evidence-Backed Problems

Rank	Problem	Evidence	Priority	Decision
1	Customer traceability gap	135,080 rows missing CustomerID	P0	Protect customer attribution and retention reporting.
2	Invalid quantity states	10,624 rows with Quantity <= 0	P1	Separate sales from non-sale transaction states.
3	Transaction-state containing cancellation lines	9,288 cancellation lines	P0	Govern cancellations and returns separately from sales KPIs.
4	Invalid price states	2,517 rows with UnitPrice <= 0	P1	Quarantine invalid price records before publication.

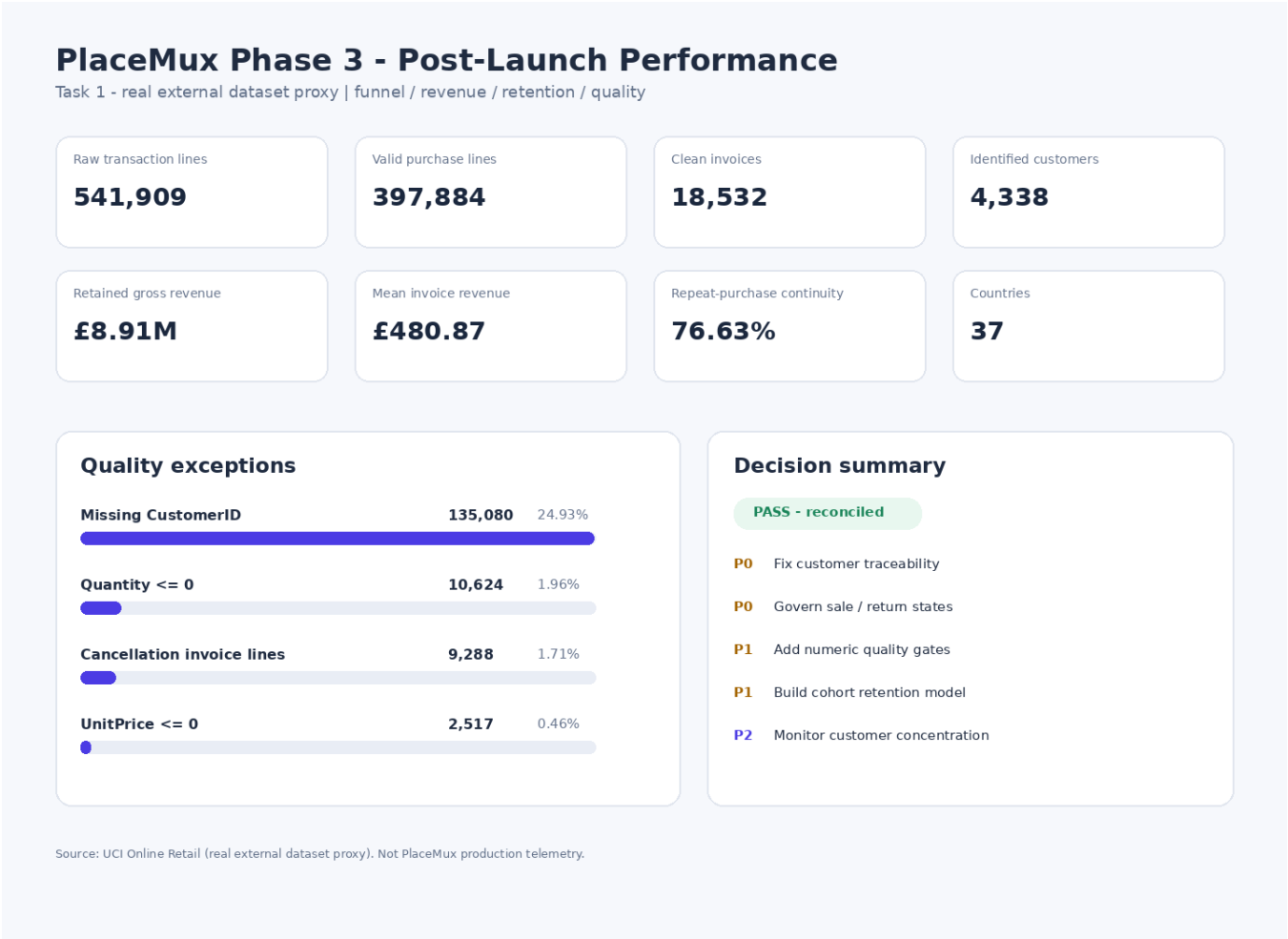
The first P0 is customer traceability because it is the largest observed source-quality gap and directly constrains customer-level retention/value reporting. The second P0 is transaction-state semantics: cancellation and return states must not silently enter sales KPIs.

4. Phase-3 Analytics Backlog

Priority	Work item	Decision changed
P0	Customer identity completeness / guest-vs-known model	Can customer retention and value reporting be trusted?
P0	Sale / cancellation / return event model	Are revenue and order KPIs aligned with commercial state?
P1	Pre-publication numeric quality gates	Should a batch publish, quarantine or require review?
P1	Cohort retention model	Which cohorts need intervention and when?
P2	Customer revenue-concentration monitoring	Is commercial risk becoming too concentrated?

5. Dashboard Evidence

This preview is generated from the same verified real-data baseline and shows the decisions surfaced by the reporting layer. Running the supplied analysis script against the local XLSX regenerates the full dashboard from source.



The HTML dashboard is supplementary evidence. The underlying calculation chain remains the Python source plus machine-readable outputs.

6. Repository Evidence & Reproducibility

Artifact	Purpose
phase3/launch_performance_analysis.py	Real-data load, validation, cleaning, calculations, rankings and dashboard generation.
phase3/launch_performance_metrics.csv	Core performance metrics.
phase3/launch_quality.csv	Source-quality exception summary.
phase3/launch_revenue_monthly.csv	Monthly revenue output.
phase3/launch_problem_rankings.csv	Problem ranking.
phase3/launch_validation.json	Schema and revenue reconciliation checks.
phase3/launch_dashboard.html	Self-contained decision dashboard.
phase3/task1_launch_performance_report.md	Definitions, source and methodology.
phase3/task1_problem_rankings.md	Ranked problems and evidence.
phase3/task1_analytics_backlog.md	Decision-linked backlog.

Reproduction command: `python phase3/launch_performance_analysis.py --source "data/online_retail/Online Retail.xlsx"`

Honesty note: this is a real external dataset substitution. The task brief asks for PlaceMux production data, but none is supplied. No synthetic PlaceMux performance numbers are presented. The same calculation contract should be rerun against actual PlaceMux event, payment, retention and quality tables when those exports become available.

UCI source: Online Retail, DOI 10.24432/C5BW33, CC BY 4.0.